Digital Data Transmission 2026–27

Universitat Pompeu Fabra Barcelona

Seminar 7. Intersymbol interference in PAM

A transmitter sends bits with the 4-PAM constellation below, with $a > 0$ and the labels shown, two bits per symbol.



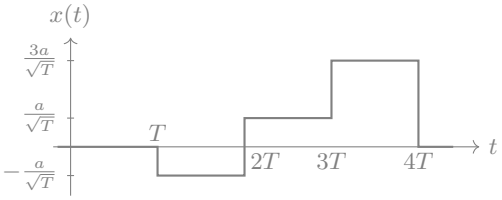
The symbols x_1, x_2, x_3 are sent as $x(t) = \sum_{i=1}^{3} x_i\, \phi(t - iT)$, with the rectangular pulse $\phi(t) = 1/\sqrt{T}$ on $[0, T)$ and zero elsewhere, and period $T > 0$, so the symbol x_i occupies $[iT, (i+1)T)$. The receiver filters with $h(t) = \phi(-t)$. By Section 5.3 of the notes, without noise the filter output is $q(t) = \sum_{k=1}^{3} x_k\, g(t - kT)$, where $g(\tau) = 1 - |\tau|/T$ for $|\tau| \le T$ and $g(\tau) = 0$ otherwise. The samples go to the nearest-point detector of 4-PAM. The six bits to send are 01 11 10, and there is no additive noise in questions 4 to 6.

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1. Which symbols (x_1, x_2, x_3) carry the six bits? Draw the transmitted waveform $x(t)$ and mark its levels.

(a) $(-3a, a, 3a)$.
(b) $(-a, 3a, a)$.
(c) $(-a, a, 3a)$. ⋆

Solution: The mapper of Section 5.1 reads the bits two at a time. With the labels of the figure, 01 is $-a$, 11 is a and 10 is $3a$, so $(x_1, x_2, x_3) = (-a, a, 3a)$. By Section 5.2 the symbol x_i occupies $[iT, (i+1)T)$ with the level $x_i/\sqrt{T}$. The answer is (c).



2. The bit rate is $R_\mathrm{b} = 6$ Mbit/s. What is the symbol period T? Obtain the symbol rate first.

(a) $\frac{1}{6}\,\mu$s.
(b) $\frac{1}{3}\,\mu$s. ⋆
(c) $\frac{1}{12}\,\mu$s.

Solution: Without coding each symbol carries $\log_2 4 = 2$ bits, so by Section 5.1

$$R_\mathrm{s} = \frac{R_\mathrm{b}}{\log_2 M} = 3 \text{ Msymbols per second}$$

and $T = 1/R_\mathrm{s} = \frac{1}{3}\,\mu$s. The answer is (b).

3. What is the energy per bit E_b of this constellation, with equiprobable points? Calculate the energy per symbol E_s first.

(a) $\frac{5}{2}a^2$. ⋆
(b) $5a^2$.
(c) $10a^2$.

Solution: The four points have squared values $9a^2$, a^2, a^2 and $9a^2$, so by Section 5.1

$$E_\mathrm{s} = \tfrac{1}{4}\left(9a^2 + a^2 + a^2 + 9a^2\right) = 5a^2.$$

Each symbol carries two bits, so $E_\mathrm{b} = E_\mathrm{s}/2 = \frac{5}{2}a^2$. The answer is (a).

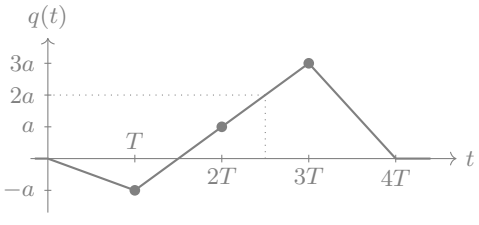
4. What is the filter output at $t = \frac{5T}{2}$, halfway between the instants of the second and the third symbols? Draw $q(t)$ for $0 \le t \le 4T$.

(a) a.
(b) $4a$.
(c) $2a$. ⋆

Solution: In $q(t) = \sum_k x_k\, g(t - kT)$ the triangle of the second symbol is evaluated at $\frac{T}{2}$ and that of the third at $-\frac{T}{2}$, where $g = \frac{1}{2}$, and the triangle of the first symbol is zero there. Hence

$$q\left(\tfrac{5T}{2}\right) = \tfrac{1}{2}x_2 + \tfrac{1}{2}x_3 = \tfrac{1}{2}a + \tfrac{3}{2}a = 2a.$$

The sum of the three triangles joins the points (kT, x_k) by straight lines, from $q(0) = 0$ to $q(4T) = 0$. The answer is (c).



5. The receiver samples too early, at $t = iT - \frac{T}{4}$. What are the three samples? Check whether the three decisions are still correct.

(a) $-\frac{1}{2}a$, $\frac{3}{2}a$ and $\frac{9}{4}a$.
(b) $-\frac{3}{4}a$, $\frac{1}{2}a$ and $\frac{5}{2}a$. ⋆
(c) $-\frac{3}{4}a$, $\frac{3}{4}a$ and $\frac{9}{4}a$.

Solution: At $t = iT - \frac{T}{4}$ the triangle of the symbol x_i is evaluated at $-\frac{T}{4}$, where $g = \frac{3}{4}$, and the triangle of the previous symbol at $\frac{3T}{4}$, where $g = \frac{1}{4}$. The next symbol does not contribute, because $g(-\frac{5T}{4}) = 0$. Hence

$$q\big(iT - \tfrac{T}{4}\big) = \tfrac{3}{4}x_i + \tfrac{1}{4}x_{i-1}, \qquad x_0 = 0.$$

The samples are $-\frac{3}{4}a$, $\frac{3}{4}a - \frac{1}{4}a = \frac{1}{2}a$ and $\frac{9}{4}a + \frac{1}{4}a = \frac{5}{2}a$. With the thresholds 0 and $\pm 2a$ the decisions are $-a$, a and $3a$, all correct. The second term is the intersymbol interference of Section 5.3. The answer is (b).

6. The receiver now samples much earlier, at $t = iT - \frac{3T}{4}$. How many of the three decisions are wrong?

(a) Two. ⋆
(b) One.
(c) None.

Solution: As in question 5, now with $g(-\frac{3T}{4}) = \frac{1}{4}$ for the symbol x_i and $g(\frac{T}{4}) = \frac{3}{4}$ for the previous one,

$$q\big(iT - \tfrac{3T}{4}\big) = \tfrac{1}{4}x_i + \tfrac{3}{4}x_{i-1}, \qquad x_0 = 0.$$

The samples are $-\frac{1}{4}a$, $\frac{1}{4}a - \frac{3}{4}a = -\frac{1}{2}a$ and $\frac{3}{4}a + \frac{3}{4}a = \frac{3}{2}a$. With the thresholds 0 and $\pm 2a$ the decisions are $-a$, $-a$ and a, while $-a$, a and $3a$ were sent: the second and the third are wrong. In the second and the third samples the previous symbol weighs $\frac{3}{4}$ and the own symbol $\frac{1}{4}$; the first is only attenuated, since $x_0 = 0$. The intersymbol interference causes errors without any noise. The answer is (a).

7. The same six bits can be sent with QPSK, which puts one bit on a cosine and one on a sine in each period, each as $\pm a$ (Section 5.4). Both systems send two bits per period with minimum distance $2a$. With equiprobable points in both constellations, what is the ratio between the average energy per bit of 4-PAM, from question 3, and that of QPSK?

(a) 5.
(b) $\frac{5}{2}$. ⋆
(c) $\frac{5}{4}$.

Solution: A QPSK symbol has the two coordinates $\pm a$, so its energy is $E_\mathrm{s} = a^2 + a^2 = 2a^2$ for every point, and with two bits per symbol $E_\mathrm{b} = a^2$. By question 3, 4-PAM has $E_\mathrm{b} = \frac{5}{2}a^2$, so the ratio is

$$\frac{\frac{5}{2}a^2}{a^2} = \frac{5}{2},$$

about 4 dB. For the same bit rate and the same minimum distance, a second coordinate on the sine costs less energy than two more levels on the same coordinate. The answer is (b).